

Mojtaba Almadan

Computer & Information Technology Engineer

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mjs19999.github.io/portfolio

SUMMARY

Computer Engineer skilled in **robotics, AI, and full-stack** system design. Experienced in developing production-grade solutions from **IoT and embedded systems** to **robotic simulation and control**. Proficient in **Python, machine learning, and scalable software infrastructure** for real-world applications.

PROFESSIONAL EXPERIENCE

Computer Engineer – Systems & Internal Tools

Jul 2023 – Present

Khonaini International Company Ltd. – Jubail

- Designed and implemented an internal asset and employee management system in **Django and PostgreSQL**, replacing legacy enterprise software and manual datasheet workflows; it now tracks 684 assets, 933 employee records and 1,423 custody assignments.
- Modelled equipment custody as an append-only ledger with dual on-screen signature capture on every transfer and **role-based permissions** across four groups, ensuring **data consistency** and a provable chain of possession.
- Built **backup and restore** management commands covering database, media and source, and recovered from a failed restoration by resolving content-type and permission collisions.
- Provisioned and configured endpoints (**OS image, software, drivers**), resolved **network/server** connectivity issues, and created user documentation and training materials.

AI Integration Engineer

Jun 2021 – Jan 2023

UAS2030 – Dhahran

- Developed an **AI system for object identification, recognition, and tracking** in **collaboration** with **computer vision** and **machine learning** specialists.
- Conducted tests and experiments to evaluate system reliability and performance in different environments and scenarios.
- Integrated AI components into a **larger system pipeline**, demonstrating strong **attention to detail** and effective **teamwork** in a demanding and collaborative environment.

EDUCATION

Master of Science, Internet of Things

2025 – Present

King Fahd University of Petroleum & Minerals (KFUPM) – Dhahran

Transducers, IoT Gateways, IoT Platforms, LPWAN, MQTT, Zigbee, LoRaWAN, Edge Computing, Telematics, Real-Time Analytics, IIoT, Smart Home, Digital Transformation, Interoperability

Bachelor of Science, Computer Engineering

2017 – 2023

King Fahd University of Petroleum & Minerals (KFUPM) – Dhahran

Computer Architecture, Digital Design, Microprocessors, Signal Processing, Control Systems, Operating Systems, Networking

KEY SKILLS

Backend & Web – Full-stack, Python, Django, Django REST Framework, RESTful APIs, SQL, PostgreSQL, Database Design

Desktop & Embedded

C/C++, ESP32, FreeRTOS, Verilog, Logisim, Assembly, Java, Microsoft Office

AI & Computer Vision – OpenCV, Object Detection, Model Integration, scikit-learn, pandas

Soft Skills – Structured Problem-solving, Technical Communication, Cross-functional Collaboration, Adaptability & Learning Agility, Priority Management, Attention to Detail

ACTIVITIES

- Participated in AUVSI SUAS competition and designed a robust AI/ML model for SUAS.
- Built a 32-bit RISC processor in Logisim, single-cycle and pipelined, validated by running a selection sort in assembly – github.com/mjs19999/32-bit-cpu-in-logisim
- Implemented AES encryption/decryption as an algorithmic state machine in Verilog; 6 stars and 4 forks – github.com/mjs19999/AES_in_verilog